

T3 matched EFT Lagrangian terms: EFT_2_after_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1, S2. Remaining active T3 fields: none.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1 – S_2 contraction
$\lambda_{S_1}^{(A)}$	additional independent S_1 self-contraction
$\lambda_{S_2}^{(A)}$	additional independent S_2 self-contraction
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$H \ell e$	$F H \ell$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\sqrt{3}\epsilon M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$
								$+\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\sqrt{3}M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{\sqrt{3}M_F}{\sqrt{3}M_F}$
								$+\frac{2h\overline{\lambda_5 L_{a_3,i_1}}}{\sqrt{3}\epsilon M_F} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}}{\sqrt{3}M_F} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$
cont.								$+\frac{2h\overline{\lambda_5 L_{a_3,i_1}}}{\sqrt{3}M_F} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{4\epsilon} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{4\epsilon} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{\epsilon M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}}}{M_F}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}$	$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_3,r_1}L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{M_F}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{\lambda_5 L_{a_3,i_1}}}{\epsilon M_F} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}}{M_F} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\overline{\lambda_5 L_{a_3,i_1}}}{M_F} C P_R \overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger$	$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{4\epsilon} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
cont.									$+\frac{3}{8}\sqrt{3}h\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{4\epsilon} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}\sqrt{3}h\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}} \overline{e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$FH\ell e$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$